

STAT 525

Chapter 16

Single-Factor Studies

Categorical

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One-Way ANOVA

- Response variable Y is again continuous
- Explanatory variable is categorical
 - Often called a factor
 - The possible values are its levels
- A generalization of the independent two-sample t-test (i.e., can be used when there are more than two levels)

ANOVA vs. Regression

- One-way ANOVA a special case of regression using indicator variables
- Recall in comparing regression lines, indicator variables were used to describe differences in intercepts (i.e, means)
- Consider the linear model $Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$ involving three groups where X_1 is the indicator for group 1 and X_2 is the indicator for group 2
 - Group 1: $Y_i = \beta_0 + \beta_1 + \varepsilon_i = \mu_1 + \varepsilon_i$
 - Group 2: $Y_i = \beta_0 + \beta_2 + \varepsilon_i = \mu_2 + \varepsilon_i$
 - Group 3: $Y_i = \beta_0 + \varepsilon_i = \mu_3 + \varepsilon_i$
- Indicators remove “linear” structure among means
*Using 1 as indicator of interest,
& otherwise*

The Data / Notation

- Y is the response variable
- X is the factor with r levels. These levels are often called groups or treatments.
- Let Y_{ij} be the $\begin{matrix} \text{Trt 1: } Y_{1j} \\ \text{Trt 2: } Y_{2j} \end{matrix}$
 - j^{th} observation ($j = 1, 2, \dots, n_i$)
 - in the i^{th} group ($i = 1, 2, \dots, r$)

Example (Page 685)

- Kenton Food Company wants to test ^{$i=4$ groups} four different package designs for a new breakfast cereal
- Twenty “similar” stores were selected to be part of the experiment
- ^{$20-1=19=n$} Package designs randomly and equally assigned to stores. Fire hit one store so it was dropped
- Y is the number of cases sold
- X is the package design with $r = 4$ levels
 - $i = 1, 2, 3, 4$
 - $j = 1, 2, \dots, n_i$ where $n_i = 5, 5, 4, 5$ respectively ^{Total $n=19$}
 - will use n when n_i constant

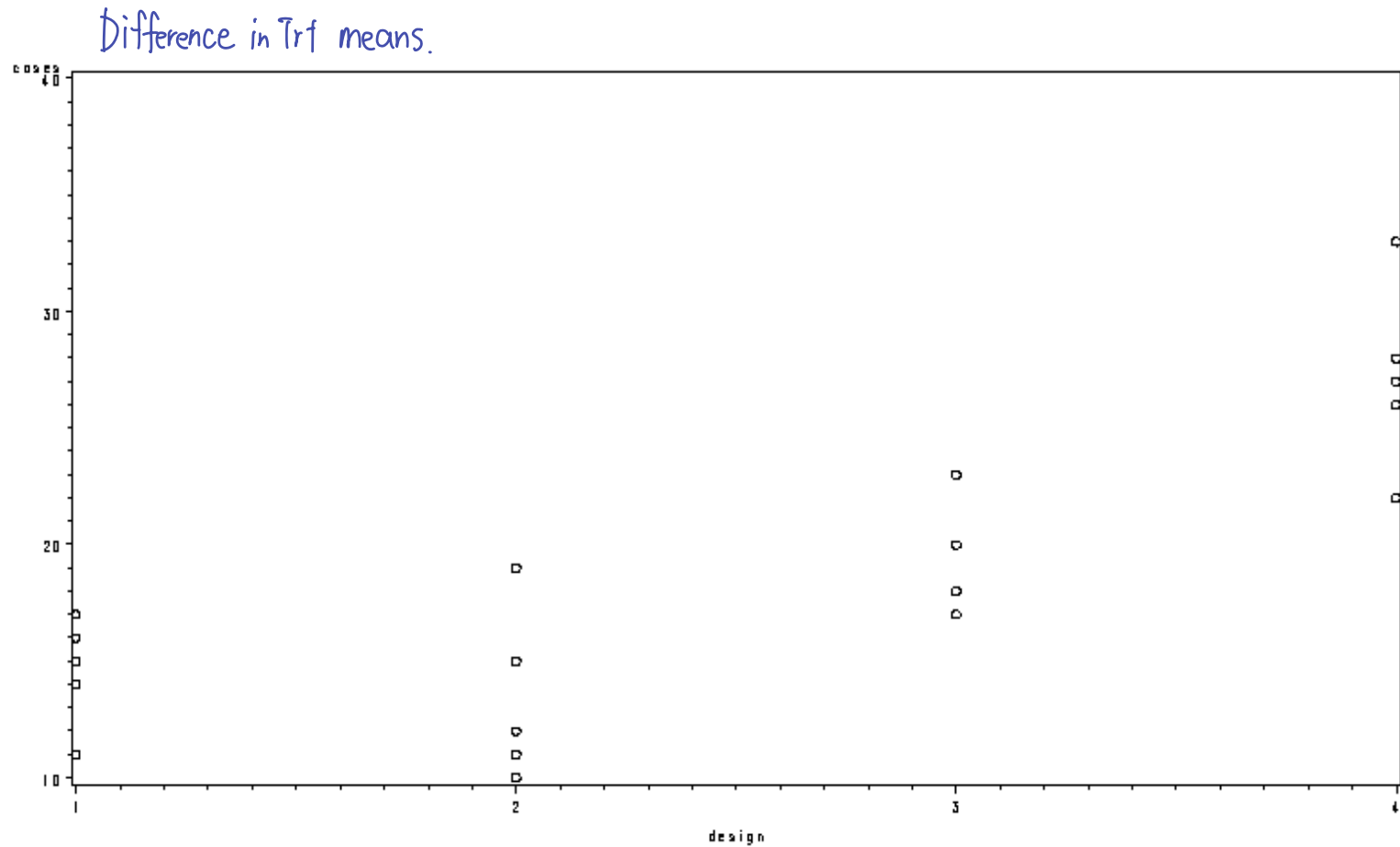
The Data

```
data a1;  
    infile 'u:\.www\datasets525\CH16TA01.TXT';  
    input cases design store;  
proc print; run; quit;
```

Obs	cases	design	store
1	11	1	1
2	17	1	2
3	16	1	3
4	14	1	4
5	15	1	5
6	12	2	1
7	10	2	2
8	15	2	3
9	19	2	4
10	11	2	5
11	23	3	1
12	20	3	2
13	18	3	3
14	17	3	4
15	27	4	1
16	33	4	2
17	22	4	3
18	26	4	4
19	28	4	5

Scatterplot

```
symbol1 v=circle i=none;  
proc gplot data=a1;  
    plot cases*design/frame;  
run; quit;
```



X-axis has no numerical meaning.

The Model

- Same assumptions as regression except for the linear relationship between X and Y
- All observations are assumed independent
- All observations are normally distributed with
 - means which may depend on the levels of the factors
 - constant variance
- Often presented in terms of cell means or factor effects

Ex): Mean of Trt 4 >>> Trt 1.

The Cell Means Model

- Expressed numerically

$$Y_{ij} = \mu_i + \varepsilon_{ij}$$

where μ_i is the theoretical mean of all observations at level i (or in cell i) Ex: 4 Trts, so 4 μ 's ($\mu_1, \mu_2, \mu_3, \mu_4$)

- The ε_{ij} are iid $N(0, \sigma^2)$ which implies the Y_{ij} are independent $N(\mu_i, \sigma^2)$

- Parameters

$$- \mu_1, \mu_2, \dots, \mu_r$$

$$- \sigma^2$$

Primary Question

- Does the explanatory variable X help explain Y ?
- Since the factor levels only affect the cell means we can similarly ask ...
- Does μ_i depend on i ?
 - $H_0 : \mu_1 = \mu_2 = \dots = \mu_r = \mu$ *H_0 : No difference across Trt means*
 - H_a : at least one μ_i different
- But this test doesn't tell you the magnitude of difference, only presence of difference.
- So further investigation is need if H_0 is rejected.

Estimates / Inference

- Derive the following result via matrix form of linear regression
- Estimate μ_i by the sample mean of the observations at level i

$\hat{\mu}_i$ for each level

$$\hat{\mu}_i = \bar{Y}_{i.}$$

- For each level i , also estimate of the variance

s_i^2 for each level

$$s_i^2 = \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2 / (n_i - 1)$$

- These s_i^2 are combined to estimate σ^2

same n of obs. in each level

← If n_i were constant, could compute s^2 by averaging the s_i^2 's

← More general formula pools s_i^2 using weights proportional to sample size (i.e., df)

If different n of obs. in each level, use s^2 weights.

$$s^2 = \frac{\sum_{i=1}^r (n_i - 1) s_i^2}{\sum_{i=1}^r (n_i - 1)} = \frac{\sum_{i=1}^r (n_i - 1) s_i^2}{n_T - r}$$

where n_T is the total number of obs

ANOVA Table

- Similar ANOVA table construction
- Plug in $\bar{Y}_{i.}$ as fitted value

Source of Variation	df	SS
Between Groups	$r - 1$	$\sum_{i=1}^r n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2$
Within Groups	$n_T - r$	$\sum_{i=1}^r \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2$
Total	$n_T - 1$	$\sum_{i=1}^r \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{..})^2$

- Note that

$$\begin{aligned} \text{Overall Mean } \bar{Y}_{..} &= \sum_{i=1}^r \sum_{j=1}^{n_i} Y_{ij} / n_T \\ &\quad \text{ } \quad \quad \quad n_{\text{Total}} \end{aligned} \qquad \begin{aligned} \text{group mean } \bar{Y}_{i.} &= \sum_{j=1}^{n_i} Y_{ij} / n_i \\ &\quad \quad \quad \quad \quad n \text{ in each group} \end{aligned}$$

- $SSM = SS(B)$, aka the between-group variation;
- $SSE = SS(W)$, aka the within-group variation.

Expected Mean Squares (EMS)

- All means squares are random variables
- Can show $E(\text{MSE}) = \sigma^2$ (page 696)
- Can also show (page 697)

$$E(\text{MSR}) = \sigma^2 + \frac{\sum n_i (\mu_i - \mu_{\cdot})^2}{r - 1}$$

where $\mu_{\cdot} = \frac{\sum n_i \mu_i}{n_T}$

- If H_0 true, MSR unbiased estimate of σ^2 . More specifically, SSE/σ^2 and SSR/σ^2 are independent χ^2 distribution
- In more complicated ANOVA models, EMS (Hasse diagram; STAT 514) guides us how to construct F tests

for Continuous: PROC REG

for both Continuous and Categorical

Example (Page 685) – Use PROC GLM in SAS

```
/* GLM: Uses least squares method to fit general linear models, and */  
/* provides regression, ANOVA, ANCOVA, MANCOVA, partial correlation */  
/* Automatically create indicator variable by class statement */  
proc glm data=a1;  
    class design;  
    model cases=design;  
    means design;  
    lsmeans design / stderr;  
run; quit;
```

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	588.2210526	196.0736842	18.59	<.0001
Error	15	158.2000000	10.5466667		
Corrected Total	18	746.4210526			

R-Square	Coeff Var	Root MSE	cases Mean
0.788055	17.43042	3.247563	18.63158

Source	DF	Type I SS	Mean Square	F Value	Pr > F
design	3	588.2210526	196.0736842	18.59	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
design	3	588.2210526	196.0736842	18.59	<.0001

The GLM Procedure

Level of design	N	-----cases-----	Mean	Std Dev
1	5	Level Means {	14.6000000	2.30217289
2	5		13.4000000	3.64691651
3	4		19.5000000	2.64575131
4	5		27.2000000	3.96232255

* only use means for equal group size and no covariates.

Least Squares Means

design	cases LSMEAN	Standard Error	Pr > t
1	14.6000000	1.4523544	<.0001
2	13.4000000	1.4523544	<.0001
3	19.5000000	1.6237816	<.0001
4	27.2000000	1.4523544	<.0001

Option	What it does	When to use	Problem
MEANS	Computes simple group averages	Only OK in balanced ANOVA with no covariates	Does not adjust for covariates or unequal sample sizes
✓ LSMEANS	Computes adjusted means (least squares means) based on the model	Best for ANCOVA, regression models, or unbalanced designs	None, this is preferred

plus some plots.

- **MEANS** only uses the observations from a specific group
 - $4 \times 2.30^2 + 4 \times 3.65^2 + 3 \times 2.65^2 + 4 \times 3.96^2 = 158.24$. Except for rounding, this is equal to SSE.
 - $19 - 4 = 15$, which is the df error in the ANOVA table.
- **LSMEANS** uses all the observations and least squares method
 - $SE_i = \sqrt{MSE/n_i}$.

Example (Page 685) – Use PROC MIXED in SAS

```
/* MIXED: generalizes the linear models in PROC GLM & fits linear mixed models */  
proc mixed data=a1;  
    class design;  
    model cases=design;  
    lsmeans design;  
run; quit;
```

- Handles random effects, longitudinal data
- Allows missing data

Type 3 Tests of Fixed Effects

Effect	Num DF	Den DF	F Value	Pr > F
design	3	15	18.59	<.0001

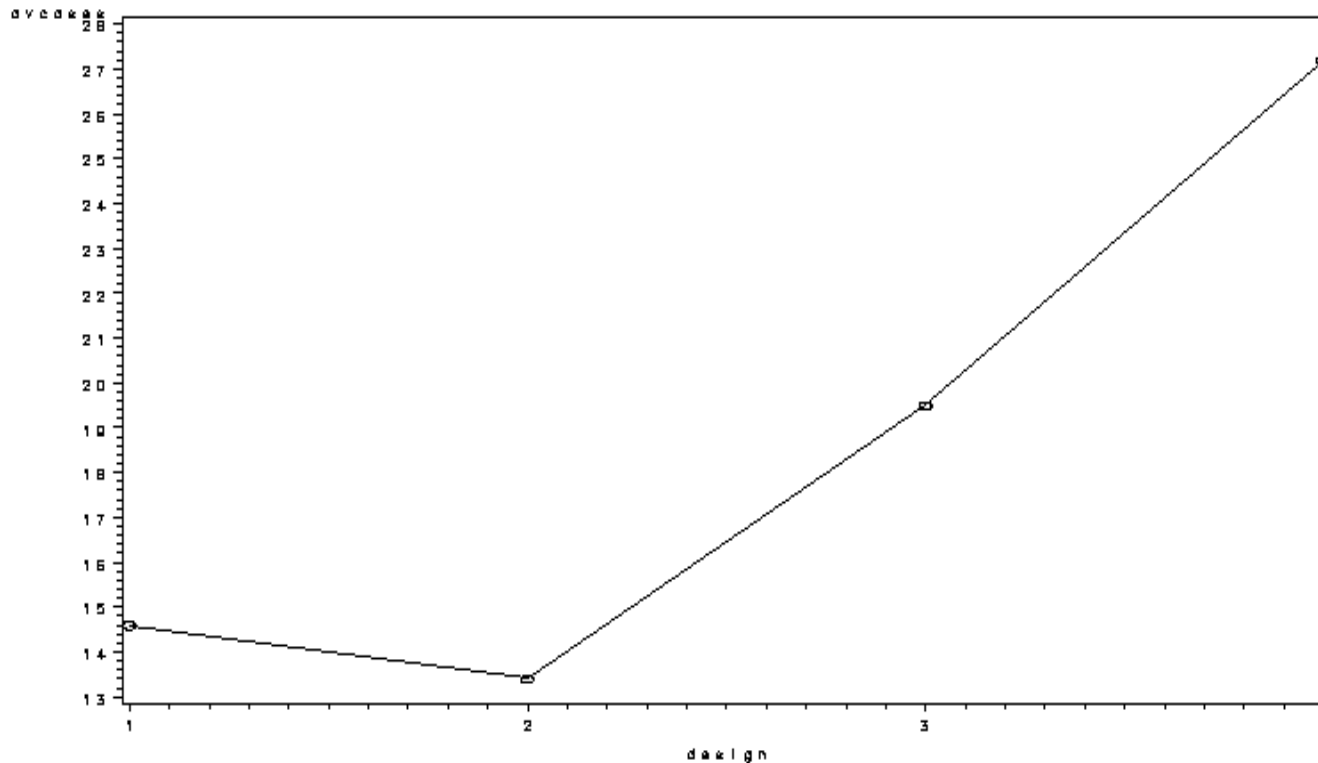
Least Squares Means

	Estimate	Standard Error	DF	t Value	Pr > t
design					
1	14.6000	1.4524	15	10.05	<.0001
2	13.4000	1.4524	15	9.23	<.0001
3	19.5000	1.6238	15	12.01	<.0001
4	27.2000	1.4524	15	18.73	<.0001

Scatterplot of Means

Generated by `lsmeans design/plot=meanplot(join);` in glm procedure, or manually:

```
proc means data=a1;  
    var cases; by design;  
    output out=a2 mean=avcases;  
symbol1 v=circle i=join;  
proc gplot data=a2;  
    plot avcases*design/frame;  
run; quit;
```



The Factor Effects Model

- A reparameterization of the cell means model
- A very useful way of looking at more complicated ANOVA models (i.e., more than one factor)
- Null hypotheses are easier to state
- Expressed numerically

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

- Parameters
 - $\tau_1, \tau_2, \dots, \tau_r$
 - μ, σ^2
- Factor effects model has $r+2$ parameters while the cell means model has $r+1$ parameters

Model Identifiability

- Consider $r = 3$ with $\mu_1 = 10, \mu_2 = 0$, and $\mu_3 = 20$
 - $\mu = 0, \tau_1 = 10, \tau_2 = 0, \tau_3 = 20$
 - $\mu = 10, \tau_1 = 0, \tau_2 = -10, \tau_3 = 10$
 - $\mu = 100, \tau_1 = -90, \tau_2 = -100, \tau_3 = -80$
- Factor effects model has non-unique solution
- Solution: put constraints on τ_i 's to reduce the parameters number by 1
- Examples of constraints
 - $\tau_r = 0$ (SAS approach)
 - $\sum \tau_i = 0$ (conceptual approach)
- Constraints get a bit more complicated when n_i not constant (pages 709-710) but with same concept

Consequences of Constraints

- Consider $r = 3$ with $n_i = n$
- Factor effects model with $\sum \tau_i = 0$

$$\begin{aligned}E(\bar{Y}_{..}) &= \frac{3\mu + \sum \tau_i}{3} = \mu \\E(\bar{Y}_{i.}) &= \mu + \tau_i\end{aligned}$$

- μ is the grand mean
 - τ_i is the effect of the i^{th} factor
- Factor effects model with $\tau_r = 0$

$$\begin{aligned}E(\bar{Y}_{3.}) &= \mu \\E(\bar{Y}_{1.} - \bar{Y}_{3.}) &= \mu + \tau_1 - \mu = \tau_1\end{aligned}$$

- μ is the mean of the r^{th} group
- τ_i is the difference between the means of group i and group r

- Different constraints result in different parameter / parameter estimates
- Many estimates, however, are the same regardless of constraint
 - $\hat{\mu} + \hat{\tau}_1 = \text{trt 1 mean}$
 - $\hat{\mu} + \hat{\tau}_3 = \text{trt 3 mean}$
 - $\hat{\tau}_1 - \hat{\tau}_3 = \text{difference in trt 1 and trt 3}$
 - $\hat{\tau}_1 - \hat{\tau}_2 = \text{difference in trt 1 and trt 2}$
- These are primarily the ones of interest

Hypotheses

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_r = \mu$$

H_a : at least one μ_i different

is translated into

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_r = 0$$

H_a : at least one $\tau_i \neq 0$

Regression Approach

- We can use multiple regression to produce results based on the factor effects model

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$$

- Consider the restriction $\sum \tau_i = 0$
- Because of this restriction, effectively there are $r - 1$ regression coefficients /parameters

$$\sum \tau_i = 0 \rightarrow \tau_r = -\tau_1 - \tau_2 - \dots - \tau_{r-1}$$

- Define k -th indicator variable ($k = 1, 2, \dots, r - 1$)

$$X_{ijk} = \begin{cases} 1, & \text{factor level at } k, \text{ i.e., } i = k \\ -1, & \text{factor level at } r, \text{ i.e., } i = r \\ 0, & \text{otherwise} \end{cases}$$

- Multiple regression model

$$Y_{ij} = \beta_0 + \beta_1 X_{ij1} + \beta_2 X_{ij2} + \dots + \beta_{r-1} X_{ij,r-1} + \varepsilon_{ij}$$

- For level i ($1 \leq i \leq r-1$)

$$Y_{ij} = \beta_0 + \beta_i + \varepsilon_{ij}$$

- For level r

$$Y_{ij} = \beta_0 - \beta_1 - \beta_2 - \dots - \beta_{r-1} + \varepsilon_{ij}$$

- Perfectly match $\mu = \beta_0$ and $\tau_i = \beta_i$ ($1 \leq i \leq r-1$)
- Solve all β_i via the multiple linear regression approach.
- $\hat{\mu} = b_0 = \sum_{i=1}^r \bar{Y}_{i.}/r$ (if n_i 's are not constant, then $b_0 \neq \bar{Y}_{..}$),
 $\hat{\tau}_i = b_i = \bar{Y}_{i.} - b_0$.


```

/* Code Indicator Variables */
data a1; set a1;
  x1=(design eq 1)-(design eq 4);
  x2=(design eq 2)-(design eq 4);
  x3=(design eq 3)-(design eq 4);
proc print data=a1; run; quit;

```

Obs	cases	design	store	x1	x2	x3
1	11	1	1	1	0	0
2	17	1	2	1	0	0
3	16	1	3	1	0	0
4	14	1	4	1	0	0
5	15	1	5	1	0	0
6	12	2	1	0	1	0
7	10	2	2	0	1	0
8	15	2	3	0	1	0
9	19	2	4	0	1	0
10	11	2	5	0	1	0
11	23	3	1	0	0	1
12	20	3	2	0	0	1
13	18	3	3	0	0	1
14	17	3	4	0	0	1
15	27	4	1	-1	-1	-1
16	33	4	2	-1	-1	-1
17	22	4	3	-1	-1	-1
18	26	4	4	-1	-1	-1
19	28	4	5	-1	-1	-1

```
proc reg data=a1;
  model cases=x1 x2 x3;
run; quit;
```

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	588.22105	196.07368	18.59	<.0001
Error	15	158.20000	10.54667		
Corrected Total	18	746.42105			
Root MSE		3.24756	R-Square	0.7881	
Dependent Mean		18.63158	Adj R-Sq	0.7457	
Coeff Var		17.43042			

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	18.67500	0.74853	24.95	<.0001
x1	1	-4.07500	1.27081	-3.21	0.0059
x2	1	-5.27500	1.27081	-4.15	0.0009
x3	1	0.82500	1.37063	0.60	0.5562

- The mean of the means is 18.675
- The treatment means are $18.675 - 4.075 = 14.6$, $18.675 - 5.275 = 13.4$, $18.675 + 0.825 = 19.5$, and $18.675 + 4.075 + 5.275 - 0.825 = 27.2$
- The same output as from PROC GLM before

SAS Regression Approach

- `class` statement constructs the following r indicator variables

$$X_{ijk} = \begin{cases} 1 & \text{if } i = k \\ 0 & \text{otherwise} \end{cases}$$

- Because of the intercept (column of 1's) there is complete dependence ($\mathbf{X}'\mathbf{X}$ doesn't have an inverse)

$$\mathbf{1} = \mathbf{X}_1 + \mathbf{X}_2 + \dots + \mathbf{X}_r$$

- SAS computes *generalized inverse* in its place. (Generalized inverse of $\mathbf{A}$, $\mathbf{A}^-$, satisfies $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$)
- $\mathbf{b} = (\mathbf{X}'\mathbf{X})^-(\mathbf{X}'\mathbf{Y})$ satisfies $(\mathbf{X}'\mathbf{X})\mathbf{b} = (\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^-(\mathbf{X}'\mathbf{Y}) = (\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^-(\mathbf{X}'\mathbf{X})\tilde{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})\tilde{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$, for some $\tilde{\boldsymbol{\beta}}$.
- Generalized inverse is not unique, and SAS choose the a particular one such that $\hat{\tau}_r = 0$.

Example (Page 685)

```
proc glm data=a1;  
  class design;  
  model cases=design / xpx inverse solution;  
run; quit;
```

The X'X Matrix

	Int	d1	d2	d3	d4	cases
Int	19	5	5	4	5	354
d1	5	5	0	0	0	73
d2	5	0	5	0	0	67
d3	4	0	0	4	0	78
d4	5	0	0	0	5	136
cases	354	73	67	78	136	7342

X'X Generalized Inverse (g2)

	Int	d1	d2	d3	d4	cases
Int	0.2	-0.2	-0.2	-0.2	0	27.2
d1	-0.2	0.4	0.2	0.2	0	-12.6
d2	-0.2	0.2	0.4	0.2	0	-13.8
d3	-0.2	0.2	0.2	0.45	0	-7.7
d4	0	0	0	0	0	0
cases	27.2	-12.6	-13.8	-7.7	0	158.2

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	588.2210526	196.0736842	18.59	<.0001
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Source	DF	Type I SS	Mean Square	F Value	Pr > F
design	3	588.2210526	196.0736842	18.59	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
design	3	588.2210526	196.0736842	18.59	<.0001

Parameter	Estimate	Standard Error	t Value	Pr > t
Intercept	27.20000000 B	1.45235441	18.73	<.0001
design 1	-12.60000000 B	2.05393930	-6.13	<.0001
design 2	-13.80000000 B	2.05393930	-6.72	<.0001
design 3	-7.70000000 B	2.17853162	-3.53	0.0030
design 4	0.00000000 B	.	.	.

NOTE: The X'X matrix has been found to be singular, and a generalized inverse was used to solve the normal equations. Terms whose estimates are followed by the letter 'B' are not uniquely estimable.

Interpretation

- Generalized Inverse Matrix of the form

$$\begin{bmatrix} (\mathbf{X}'\mathbf{X})^{-} & (\mathbf{X}'\mathbf{X})^{-}\mathbf{X}'\mathbf{Y} \\ \mathbf{Y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-} & \mathbf{Y}'\mathbf{Y} - \mathbf{Y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-}\mathbf{X}'\mathbf{Y} \end{bmatrix}$$

- Parameter estimates in upper right corner and SSE in lower right corner
- The intercept in the parameter estimation is actually the mean estimator for the last group.

Chapter Review

- One Way ANOVA
 - Cell means model
 - Factor effects model
- Regression Approach to ANOVA